## Home assignment: day 3 → day 6

Deadline: 2026-07-06, 14:59.

1. Solve the recurrence equation:
$$a_n = (n+1) + a_{n-1}, \quad a_1 = 100.$$
Hint: recall that $1 + 2 + 3 + 4 + \cdots + n = \binom{n+1}{2} = \frac{n(n+1)}{2}$.

2. Solve the reccurence equation:
$$a_n + 5a_{n-1} + 6a_{n-2} = 0, \quad a_0 = 3, a_1 = 5.$$

3. Solve the reccurence equation:
$$c_n = 2 \cdot c_{n-1} + 3, \quad c_1 = 20.$$

4. Solve the reccurence equation:
$$a_n - 7a_{n-1} + 12a_{n-2} = 6, \quad a_0 = 1, a_1 = 12.$$

5. Watch a Veritasium video «This equation will change how you see the world»: https://youtu.be/ovJcsL7vyrk?si=kh8wjM2e6dgLSXzj.

## Demo for the quiz on day 4

1. Solve the recurrence equation:
$$b_n = \frac{n+5}{n+4} b_{n-1}, \quad b_1 = 100.$$

2. Here $L$ is the lag operator and $x_n = 6^n$. Find the values
$$(1 - 5L)x_n, \quad (1 - 6L)x_n, \quad (1 - 6L)(1 - 5L)x_n.$$